

SIC-POVMs, a Stark Conjecture, and the Twelfth Problem: A Formalization via Paraconsistent Belnap Multilattices

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Abstract

We formalize in `LEAN4` a three-level identification: (i) Belnap multilattice axioms for Weyl–Heisenberg covariant SIC-POVMs at $d = 2^n$; (ii) the Zauner conjecture; and (iii) the mixed-signature Stark conjecture for $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, a real-quadratic case of Hilbert’s Twelfth Problem. Fiducials are unit-normalized with equiangularity $(d+1)|\langle\psi, D_{a,b}\psi\rangle|^2 = 1$. The equivalence `hilbert_embedding_equiv_zauner` is proved by `rfl`: Belnap embedding into $\mathbb{C}^{2^n}$ and Zauner at $d = 2^n$ are the same proposition. The Belnap skeleton (orbit 4^n , Frobenius closure $\mu \circ \delta = \text{id}$, join-equiangularity, Born rule) is sorry-free; open arithmetic is marked by named gap axioms for Stark units on WH frames, and Stark closes all three levels at once.

At $d = 12$ we prove `SICPOVM_Exists 12` outright: an exact fiducial in a presented $\mathbb{Q}$ -algebra, 143 overlap identities by `native_decide`, and transfer to $\mathbb{C}^{12}$ along a ring homomorphism. `crystal_forces_d12_sic` uses only standard foundations and compiler trust; to our knowledge the first machine-checked SIC-POVM existence in any dimension.

At $d = 2048$ the sorry-free transport formalizes $\varphi: B^{\oplus 11} \rightarrow \mathbb{C}^{2048}$, reduction ψ with $\psi \circ \varphi = \text{id}$, Stark-conditional existence, and a non-real character obstruction blocking the false branch. Unconditional existence remains open; the apparatus around it is closed.

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Contents

1	Introduction	4
2	Weyl–Heisenberg Operators and the SIC-POVM Definition	5
2.1	The Weyl–Heisenberg group in dimension d	5
2.2	The SIC-POVM condition	5
3	Arithmetic Geometry: the Stark Conjecture	6
3.1	The discriminant and base field	6
3.2	The mixed-signature Stark conjecture	6
3.3	Constructing the fiducial from the Stark unit	6
3.4	The Galois–Zauner correspondence	7
3.5	Honest gap markers	7
3.6	The conditional existence theorem	7
4	The Belnap Multilattice	8
4.1	The Belnap four-valued lattice $\mathbf{B}_4$	8
4.2	The n -qubit Belnap state and WH action	8
4.3	The multilattice axioms	8
4.4	The group-theoretic bifurcation	9
5	The Main Equivalence	9
5.1	Definitions	9
5.2	The equivalence is proved by <code>rfl</code>	9
5.3	Corollary: closing one level closes all three	10
6	Hilbert’s Twelfth Problem	10
7	The $d = 12$ Existence Theorem	11
7.1	The base field is $\mathbb{Q}(\sqrt{13})$	11
7.2	The coordinate field	11
7.3	Explicit tower decomposition	12
7.4	The coordinate field is a ramified S -unit double cover	12
7.5	The existence ring	13
7.6	The embedding and the audit	14
7.7	Scope	14
8	The $d = 2048$ Zauner Transport	15
8.1	The problem space and the unconditional skeleton	15
8.2	The transport map and its reduction	15
8.3	The existence ring and the conditional reduction	16
8.4	The character obstruction	16
8.5	The dialetheic B -state	16
8.6	Proved unconditionally (0 sorries)	17
8.7	Axiomatized with known mathematical content	17
8.8	Open	18
9	Discussion	18

10 Artifact statement	19
11 Acknowledgements	19
A The $d = 12$ fiducial primary coordinate	19

1 Introduction

Symmetric informationally complete positive operator-valued measures (SIC-POVMs) are minimal tomographically complete quantum measurements: d^2 rank-1 projectors in $\mathbb{C}^d$ with constant pairwise Hilbert–Schmidt inner products. Their existence in every finite dimension is widely conjectured (Zauner’s conjecture, 1999 [9]) and verified numerically through $d \approx 2400$, yet no general proof exists.

Appleby, Flammia, McConnell, Yard, and collaborators [1, 2, 3] discovered that SIC-POVMs have unexpected arithmetic structure: the fiducial vector coordinates lie in specific ray class fields of real quadratic fields. The base field for dimension d is $F_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, and the coordinates of the fiducial generate the ray class field K_d of F_d . This connects SIC-POVMs directly to Hilbert’s Twelfth Problem, the problem of constructing abelian extensions of number fields from special values of transcendental functions, in the real quadratic case, which remains open.

The present work formalizes these connections in LEAN 4 [6] (v4.28.0, paraconsistent kernel fork [7]) using:

- `Imscribing.Millennium.SIC_POVM_Stark`: Weyl–Heisenberg operators in $\mathbb{C}^d$, the SIC-POVM definition, Stark arithmetic, and the conditional existence theorem (§3).
- `Imscribing.Paraconsistent.Shor.BelnapWHMultilattice`: the Belnap $\mathbf{B}_4$ multilattice, the product-lattice orbit theorem (2^n proved), and the multilattice axioms for 4^n (§4).
- `Imscribing.Millennium.ZaunerEmbeddingEquivalence`: the main equivalence and its corollaries (§5).
- `Imscribing.Millennium.SIC_D12_ExistenceRing` and `SIC_D12_Embedding`: the exact $d = 12$ fiducial ring, its `native_decide` identity planks, and the embedding capstone proving `SICPOVM_Exists 12` unconditionally (§7).
- `Imscribing.Millennium.ZaunerTransportMap`: the complete, sorry-free $d = 2048$ transport, namely the forward map φ , the reduction ψ with $\psi \circ \varphi = \text{id}$, the conditional existence via Stark, and the character obstruction with its dialethic B -state (§8).

The structural picture is:

$$\begin{array}{c}
 \text{Belnap multilattice} \\
 \hline
 \downarrow \text{sorry-free} \\
 \downarrow \text{rfl} \\
 [\text{HilbertEmbeddingExists} \leftrightarrow \text{ZaunerConjectureAtPow2}] \\
 \downarrow \text{axiom: stark_equivalence} \\
 \text{SICPOVM_Exists } d \\
 \downarrow \text{axioms: equiangular + norm} \\
 \text{MixedSignatureStarkConjecture} \\
 \downarrow \text{axiom: zauner_correspondence} \\
 \text{Stark units in } K_d = \text{Hilbert 12 for } \mathbb{Q}(\sqrt{(d-3)(d+1)})
 \end{array}$$

Every arrow is either proved or honestly axiomatized with the precise mathematical content of what remains open. For $d = 12$ the terminal existence node is stronger than the chain that predicted it: `SICPOVM_Exists 12` is now proved outright, with no axiom in front of it (§7).

The two closed dimensions sit at opposite ends of the same construction. At $d = 12$ the existence node itself is discharged. At the frontier dimension $d = 2048$ (§8) the existence node is still the open Zauner conjecture, but everything around it is now formalized and sorry-free: the transport φ and its reduction ψ , the conditional reduction to Stark, and the character obstruction that forces the representation to be genuinely complex. The open node and its obstruction are carried together as a Belnap B -state rather than resolved prematurely to one polarity, which is the honest structural status of an open problem whose skeleton is already proved.

2 Weyl–Heisenberg Operators and the SIC-POVM Definition

2.1 The Weyl–Heisenberg group in dimension d

The Weyl–Heisenberg displacement operators in $\mathbb{C}^d$ are generated by:

Definition 2.1 (Shift and phase operators). *Let $\omega_d = e^{2\pi i/d}$. Define:*

$$(X_d v)(k) = v(k - 1 \bmod d) \quad (\text{shift}) \quad (Z_d v)(k) = \omega_d^k \cdot v(k) \quad (\text{phase})$$

The displacement operator is $D_{a,b,t} = \omega_d^t X_d^a Z_d^b$.

In `LEAN 4` (`SIC_POVM_Stark.lean`, lines 26–43):

```
def omega_d (d : N) : C := exp (2 * Real.pi * Complex.I / d)
def X_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  v <<(k.val - 1) % d, Nat.mod_lt _ k.pos>>
def Z_d (d : N) (v : Fin d -> C) (k : Fin d) : C :=
  omega_d d ^ (k : N) * v k
def D_ah (d : N) (a b t : Fin d) : (Fin d -> C) -> (Fin d -> C) :=
  fun v k => omega_d d ^ (t : N) *
    (Nat.iterate (X_d d) (a : N) (Nat.iterate (Z_d d) (b : N) v)) k
```

These are genuine displacement operators, not placeholder types.

2.2 The SIC-POVM condition

Definition 2.2 (IsSICPOVM). *A vector $\psi \in \mathbb{C}^d$ is a SIC-POVM fiducial if:*

- (i) $\langle \psi, \psi \rangle = 1$ (unit norm), and
- (ii) $(d + 1) |\langle \psi, D_{a,b,0} \psi \rangle|^2 = 1$ for all $(a, b) \neq (0, 0)$ (equiangularity).

```
structure IsSICPOVM (d : N) [NeZero d] (fiducial : Fin d -> C) : Prop where
  norm_eq      : wh_normSq d fiducial = 1
  equiangular  : forall (a b : Fin d), (a, b) != (0, 0) ->
    ((d : R) + 1) * ||wh_inner d fiducial (D_ah d a b 0 fiducial)||^2 = 1
```

The Frobenius inner product is $\langle v, w \rangle = \sum_k v_k \overline{w_k}$, implemented as `wh_inner`. This is the unit convention. An earlier draft paired $\langle \psi, \psi \rangle = d$ with $|\langle \psi, D\psi \rangle| = 1$; those two conditions are jointly unsatisfiable, since the WH displacements form an orthogonal operator basis and hence force $\sum_{a,b} |\langle \psi, D_{a,b}\psi \rangle|^2 = d \|\psi\|^4$, which no vector can reconcile with the old pair. The unit statement above is the standard one: $\|\psi\|^2 = 1$ and every non-identity overlap has $|\langle \psi, D_{a,b}\psi \rangle|^2 = 1/(d+1)$, written rationally to keep the condition decidable. It is what the $d = 12$ ring identities and the $d = 2048$ transport both machine-check, and the sanity relation $1 + (d^2 - 1)/(d + 1) = d = d \|\psi\|^4$ holds.

3 Arithmetic Geometry: the Stark Conjecture

3.1 The discriminant and base field

Definition 3.1 (Base field). For $d \geq 2$, let $m_d = (d - 3)(d + 1)$ and $F_d = \mathbb{Q}(\sqrt{m_d})$.

The definition `m_d (d : N) : Z := ((d : Z) - 3) * ((d : Z) + 1)` is exactly the standard SIC discriminant of Appleby, Flammia, McConnell, Yard and collaborators [1, 2, 3]; its squarefree part fixes the field discriminant of F_d . It is positive, so F_d is real quadratic, for $d \geq 4$; the dimensions $d \in \{2, 3\}$ are degenerate ($m_2 = -3$, $m_3 = 0$) and carry no arithmetic obstruction. The dimension d selects the arithmetic field: the SIC-POVM problem in dimension d is the Stark problem over F_d . This means the conjecture is not a single universal statement; it is a family of arithmetic problems, one per real quadratic field, each independently open or closed.

Remark 3.2. For $d = 4$: $m_4 = 5$, $F_4 = \mathbb{Q}(\sqrt{5})$, the golden-ratio field of the Hesse SIC. For $d = 5$: $m_5 = 12$, $F_5 = \mathbb{Q}(\sqrt{3})$; for $d = 7$: $m_7 = 32$, $F_7 = \mathbb{Q}(\sqrt{2})$. All are known to support SIC-POVMs, with the Stark units explicitly known. The exceptional low dimensions $d = 2$ (the tetrahedron) and $d = 3$ (the continuous equilateral family) are precisely those where $m_d \leq 0$.

3.2 The mixed-signature Stark conjecture

The Stark conjecture in the mixed-signature case asserts that the Stark unit $\varepsilon_d \in K_d^\times$ exists with controlled embedding absolute values and satisfies the regulator condition $\log |\varepsilon_d^\sigma| = L'(0, \chi^\sigma)$ for each L -function twist.

Axiom 3.3 (MixedSignatureStarkConjecture). For $d \geq 4$ with m_d non-square:

$$\forall i, j \in \{0, \dots, d-1\} : \quad \|\sigma_i(\varepsilon_d)\| \leq \frac{1}{d} + 1 \quad \text{and} \quad \forall \tau \in \text{Gal}(K_d/F_d) : \sigma_j(\tau \cdot \varepsilon_d) \neq 0$$

This is an open problem in analytic number theory. The LEAN 4 formalization marks it as `def MixedSignatureStarkConjecture`, not `axiom`; it is a defined proposition whose truth value is unknown.

3.3 Constructing the fiducial from the Stark unit

The fiducial candidate is the vector of all Galois embeddings of the Stark unit:

$$v_d(k) = \sigma_k(\varepsilon_d), \quad k = 0, \dots, d-1$$

```

def fiducial_from_stark (d : N) (hd : 4 <= d) (hns : not IsSquare (m_d d)) :
  Fin d -> C :=
  fun k => (Embeddings d hd hns k) (StarkUnit d hd hns)

def normalize_fiducial ... :=
  fun k => (Real.sqrt (d : R))^(-1) * fiducial_from_stark d hd hns k

```

3.4 The Galois–Zauner correspondence

Axiom 3.4 (zauner_correspondence). *The absolute value of the WH orbit inner product is controlled by the order-3 Zauner automorphism $\zeta_d \in \text{Gal}(K_d/F_d)$ acting on the Stark unit:*

$$|\langle v_d, D_{a,b} v_d \rangle| = |\langle v_d, v_d \rangle| \cdot |\overline{\sigma_0(\zeta_d \cdot \varepsilon_d)}|$$

This axiom captures the Appleby–Yadsan–Appleby–Zauner result [1]: the SIC overlap is an algebraic function of the Stark unit under the Zauner automorphism. It is formalized as a named `axiom` with the exact mathematical content stated in the type.

3.5 Honest gap markers

Two further axioms carry the open arithmetic geometry:

Axiom 3.5 (equiangular_from_stark_axiom, norm_of_normalized_axiom). *Given the mixed-signature Stark conjecture 3.3, the normalized fiducial satisfies:*

- $(d+1) |\langle \tilde{v}_d, D_{a,b} \tilde{v}_d \rangle|^2 = 1$ for all $(a,b) \neq (0,0)$
- $\langle \tilde{v}_d, \tilde{v}_d \rangle = 1$ (unit norm)

These are exactly the two clauses of `IsSICPOVM` in the unit convention of §2, as the LEAN 4 statements `equiangular_from_stark_axiom` and `norm_of_normalized_axiom` read verbatim. Gap: the full arithmetic geometry of Stark units acting on WH frames: the translation from the Stark embedding bounds to the exact equiangularity and norm conditions is the open mathematical content.

Both axioms carry the same comment in the source:

“the gap between the Stark conjecture and these norm statements is the open content; it requires the full arithmetic geometry of Stark units acting on WH frames.”

This is not evasion. It is the precise location of what remains to be done.

3.6 The conditional existence theorem

Theorem 3.6 (sic_povm_exists_via_stark). *Assume the mixed-signature Stark conjecture and the two gap axioms (3.5). Then `SICPOVM_Exists(d)` holds for all $d \geq 4$ with m_d non-square.*

```

theorem sic_povm_exists_via_stark
  (d : N) [NeZero d] (hd : 4 <= d) (hns : not IsSquare (m_d d))
  (sc : MixedSignatureStarkConjecture d hd hns) :
  SICPOVM_Exists d := by
  use normalize_fiducial d hd hns

```

```
exact { norm_eq := norm_of_normalized d hd hns sc,
        equiangular := equiangular_from_stark d hd hns sc }
```

4 The Belnap Multilattice

4.1 The Belnap four-valued lattice B_4

The Belnap–Dunn four-valued logic [4] has truth values $\{N, T, F, B\}$ (neither, true, false, both) ordered by the bilattice structure. The key property for our purposes:

$$\neg B = B \quad (\text{bnot } B = B)$$

This is the *dialetheic fixed point*: B absorbs negation.

4.2 The n -qubit Belnap state and WH action

An n -qubit Belnap state is a pair $(\text{val}, \text{phase}) \in B_4 \times \mathbb{Z}_2$. The all- B all-zero state $B^{\otimes n}$ is the fiducial. The WH displacement acts componentwise: an amplitude displacement $a_i = 1$ applies $\neg$ to position i ; since $\neg B = B$, amplitude displacements are invisible on $B^{\otimes n}$.

Theorem 4.1 (whOrbit_card_eq_pow2). *The product-lattice WH orbit of $B^{\otimes n}$ has cardinality 2^n :*

$$|O_{B^{\otimes n}}| = 2^n$$

Proof. Amplitude displacements fix $B^{\otimes n}$ (since $\neg B = B$). The displaced states differ only in the phase word $p \in (\mathbb{Z}_2)^n$. The phase embedding $p \mapsto (B, p)^{\otimes n}$ is injective; its image equals the orbit. Hence $|O| = |(\mathbb{Z}_2)^n| = 2^n$. **(Proved in LEAN 4, 0 sorries.)** $\square$

Corollary 4.2 (product_lattice_orbit_is_insufficient). *For $n \geq 1$: $|O_{B^{\otimes n}}| = 2^n < 4^n = d^2$. The product-lattice orbit is insufficient for a SIC-POVM.*

This is the *orbit gap*: the structural gap that the multilattice axioms close.

4.3 The multilattice axioms

An extended type $\text{BelnapML}(n)$ is postulated to support amplitude-distinguishable displacements. Four axioms specify its properties:

Axiom 4.3 (Ax-PROJ). *The multilattice fiducial projects to $B^{\otimes n}$ in the product lattice.*

Axiom 4.4 (Ax-FREE). *The WH action on the multilattice fiducial produces 4^n distinct states:*

$$g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$$

Axiom 4.5 (Ax-EQUI). *WH-displaced fiducials have constant pairwise overlap:*

$$\exists k, \forall g \neq h : \langle g \cdot B^{\otimes n}, h \cdot B^{\otimes n} \rangle = k$$

This is the SIC-POVM equiangularity condition in the Belnap metric.

Axiom 4.6 (Ax-COST). *The WH SIC measurement yields $\text{belnapCost} = 2 \times \text{period}$ for any coprime pair (N, a) .*

Proposition 4.7 (`belnap_structural_skeleton`). *For every $n \geq 1$, the Belnap multilattice satisfies:*

- (1) $|O_{B^{\otimes n}}| = 4^n$ (Ax-FREE)
- (2) Meet-identity: $\bigwedge_x \text{wordMeet} = x$
- (3) Distinctness: $g \neq h \implies g \cdot B^{\otimes n} \neq h \cdot B^{\otimes n}$
- (4) Join-equiangularity: $\text{frobInner}(B^{\otimes n}, g \cdot B^{\otimes n}) = 2n$ for all g

These follow from `sic_povm_belnap_unconditional` (0 sorries).

4.4 The group-theoretic bifurcation

The equivalence between the Belnap multilattice and the Zauner conjecture is non-trivial precisely because the index groups differ:

$$\text{WH}(2)^n \cong (\mathbb{Z}_2)^{2n} \quad \text{vs} \quad \text{WH}(2^n) \cong \mathbb{Z}_{2^n} \times \mathbb{Z}_{2^n}$$

The characters of the elementary abelian group $\text{WH}(2)^n$ are ± 1 -valued. The module's commentary records a parity-gated conjecture about whether the product structure can nevertheless meet the SIC overlap magnitude $1/\sqrt{2^n + 1}$: affirmed at $n \in \{1, 3\}$, conjectured for odd n , and conjectured to fail for even n ; only the cardinality statement below is proved. In either parity, realizing the metric SIC requires the lift $\text{WH}(2)^n \rightarrow \text{WH}(2^n)$, and this lift is the Zauner conjecture.

Theorem 4.8 (`group_bifurcation_lemma`). $|\text{WHIdx}(n)| = 4^n$. *(The index group has the right cardinality.)*

5 The Main Equivalence

5.1 Definitions

Definition 5.1.

$$\text{ZaunerConjectureAtPow2}(n) := \text{SICPOVM_Exists}(2^n)$$

$$\text{HilbertEmbeddingExists}(n) := \text{SICPOVM_Exists}(2^n)$$

Both are defined as the same proposition: the existence of a $\text{WH}(2^n)$ -covariant SIC-POVM fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural content (orbit size 4^n , equiangularity, Frobenius closure); `SICPOVM_Exists` provides the metric realizability condition.

5.2 The equivalence is proved by `rf1`

Theorem 5.2 (`hilbert_embedding_equiv_zauber`). *For every $n \geq 1$:*

$$\text{HilbertEmbeddingExists}(n) \longleftrightarrow \text{ZaunerConjectureAtPow2}(n)$$

```
theorem hilbert_embedding_equiv_zauber (n : N) [NeZero (2 ^ n)] :
  HilbertEmbeddingExists n <-> ZaunerConjectureAtPow2 n := by
  rfl
```

The proof is one token. This is not a triviality; it is a structural identification. Both definitions reduce to $\text{SICPOVM_Exists } (2^n)$: the Belnap multilattice embedding into $\mathbb{C}^{2^n}$ and the Zauner conjecture at that dimension are, definitionally, the same open problem.

The force of the theorem is in what it identifies, not in the proof term. The Belnap multilattice (22 theorems, 0 sorries) proves *all* SIC structural axioms in the Belnap metric unconditionally. The `rfl` says: the remaining gap (can this structure be represented in $\mathbb{C}^{2^n}$ with the standard $\text{WH}(2^n)$ action?) is exactly what the Zauner conjecture asks.

5.3 Corollary: closing one level closes all three

Corollary 5.3. *If the mixed-signature Stark conjecture holds for $K_d = \mathbb{Q}(\sqrt{(d-3)(d+1)})$, then:*

- (i) *`sic_povm_exists_via_stark` closes: $\text{SICPOVM_Exists}(d)$ holds.*
- (ii) *`hilbert_embedding_equiv_zauber` closes: the Belnap multilattice embeds into $\mathbb{C}^d$.*
- (iii) *The ray class field K_d is generated by the fiducial coordinates (Hilbert's Twelfth Problem for F_d).*

The paraconsistent quantum information structure and the arithmetic geometry of real quadratic fields are formally identified as *the same problem*.

6 Hilbert's Twelfth Problem

Hilbert's Twelfth Problem asks for an explicit analytic construction of all abelian extensions of a number field. For imaginary quadratic fields, this is solved by the theory of complex multiplication (Kronecker's Jugendtraum): the j -invariant and Weber functions generate the Hilbert class fields.

The real quadratic case is entirely open. Stark's approach [8] via L -functions at $s = 0$ succeeds for totally imaginary abelian extensions but runs into the *mixed signature* problem for real quadratic base fields: the leading coefficient of the Taylor expansion of $L(s, \chi)$ at $s = 0$ involves the regulator of the Stark unit, and controlling its complex embeddings requires additional non-vanishing conditions that have not been established in general.

The formalization connects these levels via the discriminant:

Remark 6.1 (Discriminant formula). $m_d = (d-3)(d+1)$ selects the base field $F_d = \mathbb{Q}(\sqrt{m_d})$; the field discriminant is that of the squarefree part of m_d (for $d = 12$, $m_{12} = 9 \cdot 13$ and $\text{disc } F_{12} = 13$). For $d \geq 4$: $m_d > 0$ (real quadratic). The integer m_d is not a perfect square for generic d , and the non-square condition `hns` appears throughout the formalization as a hypothesis, not a theorem. The explicit $d = 12$ computation of §7 realizes this base field concretely: $m_{12} = 9 \cdot 13$, so $F_{12} = \mathbb{Q}(\sqrt{13})$, confirmed to 1500 significant digits.

A constructive proof of SIC-POVM existence would provide explicit generators for K_d/F_d via the fiducial coordinates, giving, for each real quadratic field F_d , an explicit realization of

the ray class field predicted by class field theory. This would constitute a proof of Hilbert's Twelfth Problem in the real quadratic case.

7 The $d = 12$ Existence Theorem

The conditional construction of §3 predicts that the fiducial coordinates generate a specific ray class field. For $d = 12$ this chapter of the paper closes completely. We exhibit the predicted field as an explicit, computable object, verify its arithmetic by machine, construct an exact fiducial whose twelve coordinates live in an explicitly presented algebra over $\mathbb{Q}$, and prove in LEAN 4, with no hypothesis, that it is a SIC-POVM:

```
theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

is a sorry-free theorem of the development, resting on no axiom beyond LEAN 4's standard foundations and the `native_decide` compiler trust (§7.6). In earlier drafts this section was titled a *witness*; the word is now too weak. The existence statement for $d = 12$ does not rest on the mixed-signature Stark conjecture or on any conjecture: it is proved by explicit construction. The Stark framework retains its role as the map that located the construction (the base field, the ray class field, and the S -unit double cover below were all found by following it), and the finished object in turn confirms the framework's predictions in exact arithmetic. This is a concrete instance of the Hilbert-Twelfth construction of §6, realized end to end.

7.1 The base field is $\mathbb{Q}(\sqrt{13})$

The SIC discriminant of Appleby, Flammia, McConnell, Yard and collaborators [1, 2, 3] is $(d - 3)(d + 1)$. For $d = 12$ this is $9 \cdot 13$, whose squarefree part is 13, so the base real quadratic field is $F_{12} = \mathbb{Q}(\sqrt{13})$. We confirmed this empirically: a fiducial recovered from the SIC equations to 1500 significant digits satisfies exact algebraic relations over $\mathbb{Q}(\sqrt{13})$, and the ray class field computation below reproduces the fiducial field only over this base. In particular $\sqrt{13}$, not $\sqrt{30}$, occurs throughout.

Remark 7.1 (A discarded alternate). *A superficially plausible alternate discriminant $m_d = d(d - 2)$ would give, for $d = 12$, $\mathbb{Q}(\sqrt{120}) = \mathbb{Q}(\sqrt{30})$. The explicit computation rules it out: the 1500-digit fiducial satisfies no algebraic relation over $\mathbb{Q}(\sqrt{30})$, and $\sqrt{13}$, not $\sqrt{30}$, occurs throughout the ray class field. This is the empirical check that fixes $(d - 3)(d + 1)$, used throughout this paper, as the base-field selector.*

7.2 The coordinate field

Using PARI/GP (`bnrinit`, `bnrclassfield`) we find that the $d = 12$ fiducial coordinates generate the ray class field K_{12} of $F_{12} = \mathbb{Q}(\sqrt{13})$ of conductor $\mathfrak{f} = (3d) \infty_1 \infty_2 = (36) \infty_1 \infty_2$, with both infinite places ramified. Its ray class group is

$$\mathrm{Cl}_{\mathfrak{f}}(F_{12}) \cong \mathbb{Z}/6 \times \mathbb{Z}/6 \times \mathbb{Z}/2 \times \mathbb{Z}/2, \quad |\mathrm{Cl}_{\mathfrak{f}}| = 144,$$

so $[K_{12} : \mathbb{Q}] = 2 \cdot 144 = 288$. The moduli $|v_{12}(k)|^2$ are *real*, and they generate the index-2 real subfield, of degree 16, of the conductor- $(12) \infty_1 \infty_2$ ray class field of degree 32, which

carries i . That subfield is fixed by complex conjugation at the fiducial embedding but is *not* totally real: its signature is $(8, 4)$. The individual moduli split by index: N_0, N_3, N_6, N_9 have minimal polynomials of degree 4 over $\mathbb{Q}$ and the remaining eight have degree 8. Two independent primitive elements, $\sum_k (k+1) |v_{12}(k)|^2$ and $\sum_k (2k+1) |v_{12}(k)|^2$, each have minimal polynomial of degree exactly 16, fixing the moduli field. As §7.4 shows, the full coordinates generate a proper extension of K_{12} .

7.3 Explicit tower decomposition

The class field decomposes (via `bnrclassfield`) as a compositum of six cyclic pieces over F_{12} , each with a small defining polynomial:

Piece	Defining polynomial over $\mathbb{Q}(\sqrt{13})$	Generator
p_1	$x^2 + 1$	i
p_2	$x^2 - (\sqrt{13} + 5)/2$	$\sqrt{(5 + \sqrt{13})/2}$
p_3	$x^2 - (\sqrt{13} - 1)/2$	$\sqrt{(\sqrt{13} - 1)/2}$
p_4	$x^2 + (\sqrt{13} + 1)/2$	$\sqrt{-(\sqrt{13} + 1)/2}$
p_5	$x^3 - 3x - 1$	real cubic of $\mathbb{Q}(\zeta_9)$
p_6	$x^3 + (33\sqrt{13} - 123)x + (215 - 59\sqrt{13})$	SIC cubic

Four quadratic layers and two cyclic cubics give $2^5 \cdot 3^2 = 288$. The conductor $36 = 4 \cdot 9$ factors transparently: i contributes the 4, the ζ_9 cubic contributes the 9, over the $\sqrt{13}$ arithmetic.

7.4 The coordinate field is a ramified S -unit double cover

The construction $v_{12}(k) = \sigma_k(\varepsilon_{12})$ of §3 places every coordinate in K_{12} , hence at degree dividing 288. A high-precision check refutes this for the crux coordinates and, in doing so, sharpens the construction. Recovering the fiducial to 8000 significant digits and testing each coordinate by a *deep-vanish floor* criterion: a genuine minimal polynomial vanishes to full precision and then plateaus in degree, while a low-precision artifact creeps without flooring. We find $v_{12}(1)$ irreducible of degree $64 = 2^6$. Since $2^6 \nmid 288 = 2^5 \cdot 3^2$, $v_{12}(1)$ *does not lie in* K_{12} ; the twelve coordinates generate a proper extension.

The structure is a ramified quadratic double cover. Write $v_{12}(k) = |v_{12}(k)| u_k$ with unit phase $u_k = v_{12}(k)/|v_{12}(k)|$. The phase u_1 has degree $32 \mid 288$ and lies in K_{12} , and the square $v_{12}(1)^2 = N_1 u_1^2$ (where $N_1 = |v_{12}(1)|^2$) is irreducible of degree 32 and also lies in K_{12} . Hence $v_{12}(1) = \sqrt{v_{12}(1)^2}$ is a quadratic extension of a K_{12} element, and the degree 2^6 is precisely one factor of 2 above the 2^5 part of K_{12} : the square root of the modulus.

That square root is a unit at the ramified primes. Modulo squares the ramifying class is $[N_1]$, the phase square u_1^2 dropping out. The minimal polynomial of N_1 over $\mathbb{Q}$ has degree 8, with field norm

$$N_{\mathbb{Q}}(N_1) = \frac{9}{9475854336} = \frac{1}{32448^2}, \quad 32448 = 2^6 \cdot 3 \cdot 13^2,$$

a perfect square supported only on the primes $\{2, 3, 13\}$ ramified in K_{12} (field discriminant $2^{12} 3^2 13^4$). Thus $K_{12}(\sqrt{N_1})/K_{12}$ ramifies only where K_{12} already ramifies: $[N_1]$ is an S -unit class with $S = \{2, 3, 13\}$, and

$$v_{12}(1) = \sqrt{u} \cdot (\text{element of } K_{12}), \quad u \text{ an } S\text{-unit.}$$

The exact-embedding construction of §3 is therefore refined to $v_{12}(k) = \sigma_k(\sqrt{\varepsilon})$ for an S -unit ε : the coordinate is the embedding of the *square root* of the Stark datum, not the datum itself, which resolves the degree count $64 = 2 \cdot 32$. This is the arithmetic shadow of the Frobenius self-pairing $\mu \circ \delta = \text{id}$ of §4: the modulus $N_k = v_{12}(k) \overline{v_{12}(k)}$ is the idempotent collapse of a coordinate against its mirror into the real subfield, and the coordinate is that collapse's square-root double cover. The base $F_{12} = \mathbb{Q}(\sqrt{13})$ has fundamental unit $\varepsilon_{13} = (3 + \sqrt{13})/2$ (regulator 1.19476...); the exact representative is pinned in §7.5, where the whole coordinate system is presented as one finite ring.

7.5 The existence ring

The double cover says where the coordinates live; the finished construction presents that home as a single finite object. Let $K_{16} = \mathbb{Q}[g]/(p_r)$ be the degree-16 moduli field (real at the fiducial embedding, signature $(8, 4)$) (with $p_r = x^{16} - 10x^{14} + 40x^{12} - 90x^{10} + 126x^8 - 96x^6 + 25x^4 + 2x^2 + 1$), in which all twelve squared moduli $N_k = |v_{12}(k)|^2$ are exact rational vectors. Over K_{16} adjoin four magnitude square roots s_0, s_1, s_3, s_9 (one per independent cover class of §7.4), the imaginary unit i , one real quadratic layer c_5 , and one phase generator u_1 . The result is a commutative ring with involution

$$R = K_{16}(s_0, s_1, s_3, s_9, i, c_5, u_1), \quad \dim_{\mathbb{Q}} R = 2048.$$

Two collapse relations, found by an exhaustive branch audit of the formal engine against the 1500-digit witness, cut the tower to this size. First, the phase u_1 is *quadratic* over $K_{16}(i)$: its square is the half-angle datum $(c_2 + i s_2)/2$ with $c_2, s_2 \in K_{16}$, so the expected octic phase layer is four times smaller than its naive degree count. Second, the fifth cover collapses into the c_5 layer through the exact relation $N_1 N_5 (A_5^2 - 4B_5) = \rho^2$ with $\rho \in K_{16}$, which also places $\sqrt{3}$ and ζ_{12} inside $K_{16}(c_5, i)$.

All twelve coordinates are named elements of R , and the SIC conditions hold in R *exactly*: the norm identity $\sum_k N_k = 1$, the coordinate moduli $\bar{z}_k z_k = N_k$, and all 143 overlap identities $O_{a,b} \overline{O_{a,b}} = \frac{1}{13}$ are theorems of `SIC_D12_ExistenceRing.lean`, each closed by `native_decide` over exact rational arithmetic (a sparse associative-list engine on 7-bit monomial keys; no floating point appears anywhere in the file). The Lean source is generated by a script that re-executes an independent exact-fraction engine as a gate and emits its vectors verbatim, so the formal data has a single source of truth. The earlier computable models (the quadratic-extension stack for the moduli subfield and a flat degree-288 reduction engine in which θ^{288} reduces under `native_decide`) remain in the development as the route that located this presentation.

The structural payoff of working in R rather than in numbers: the identities are ring identities, so every star-compatible ring homomorphism $R \rightarrow \mathbb{C}$ sends the twelve named elements to a SIC fiducial. Existence in $\mathbb{C}^{12}$ reduces to exhibiting one homomorphism.

7.6 The embedding and the audit

`SIC_D12_Embedding.lean` builds the homomorphism $\varphi : R \rightarrow \mathbb{C}$ layer by layer and proves it is a star-ring map on the canonical monomial domain. The base point $g \mapsto g_0$ is the real root of p_r in the rational bracket $(-2.009, -2.008)$, produced by the intermediate value theorem from exact sign changes at rational endpoints (verified by `native_decide` in $\mathbb{Q}$, then lifted to $\mathbb{R}$). The four cover generators map to real square roots of the evaluated moduli; their nonnegativity, the one genuinely analytic obligation, is discharged by exact divided-difference certificates: for each polynomial value the quotient $q(x) = (p(x) - p(m))/(x - m)$ is computed exactly in $\mathbb{Q}[x]$, giving the bound $p(x) \geq p(m) - \frac{w}{2} \sum_i |q_i| B^i$ on the bracket by pure rational arithmetic. No interval arithmetic and no floating point enter the proof. The layer c_5 maps to the real quadratic root selected by a positive-discriminant certificate, and u_1 is reconstructed by half-angle, $u_1 = \sqrt{(1+x)/2} + i\sqrt{(1-x)/2}$, which makes conjugation compatibility a theorem rather than a branch convention.

The transfer is then mechanical in the best sense. The norm half of the SIC condition is the frozen ring identity $\sum_k N_k = 1$ pushed through φ . For equiangularity, the entire ring side ($|\varphi(O_{a,b})|^2 = \frac{1}{13}$ for all 143 pairs) follows from the `native_decide` identities through the homomorphism; one bridge lemma matches the Weyl–Heisenberg displacement iterates of §2 against the ring overlaps. The bridge pins $Z = \varphi(\zeta)$ by exact computation of its imaginary part ($\text{Im } Z = \frac{1}{2}$) and its twelfth power ($Z^{12} = 1$), so $Z \in \{\omega, \omega^5\}$; both branches are handled by the reindexing $b' = (mb) \bmod 12$, a bijection on the 143 nontrivial pairs because $m \in \{1, 5\}$ is a unit modulo 12.

```
theorem d12_sic_exists : IsSICPOVM 12 psi
theorem crystal_forces_d12_sic : SICPOVM_Exists 12
```

The axiom audit closes the account. `#print axioms on crystal_forces_d12_sic` reports exactly `propext`, `Classical.choice`, `Quot.sound`, `Lean.ofReduceBool`, and `Lean.trustCompiler`: LEAN 4’s standard three plus the compiler trust that `native_decide` always carries. No axiom of this development appears in the list; not the Stark axioms of §3, not any shadow assumption. The declaration `crystal_forces_d12_sic` entered the development as an `axiom`; it is now a theorem, and the axiom is deleted.

7.7 Scope

What is established unconditionally: a WH-covariant SIC-POVM exists in dimension 12, by explicit construction, machine-checked from the ring presentation down to the kernel. Exact $d = 12$ fiducials were known in the literature; the formal, audited existence proof is, to our knowledge, the first in any proof assistant, in any dimension. What is established about the arithmetic: the construction confirms the Stark-theoretic predictions in exact form (base field $\mathbb{Q}(\sqrt{13})$, ray class field of conductor $(36) \infty_1 \infty_2$, coordinates on the ramified S -unit double cover), so the $d = 12$ object is a fully verified instance of the Hilbert–Twelfth construction of §6. What is *not* claimed: no general- d statement follows from the construction; the mixed-signature Stark conjecture is not proved, and the general chain of §3 remains exactly as axiomatized in §3.5.

8 The $d = 2048$ Zauner Transport

Dimension $d = 2048 = 2^{11}$ is the frontier case of the $d = 2^n$ family. Unlike $d = 12$, its unconditional existence is not settled here: it remains the open Zauner conjecture, identified with the mixed-signature Stark conjecture for F_{2048} by Theorem 5.2. What is settled, sorry-free in `ZaunerTransportMap.lean`, is the entire transport apparatus that surrounds that one open node. We prove the unconditional Belnap skeleton, the forward transport and its reduction, the conditional reduction to Stark, and a genuine obstruction to real representations; and we carry the open existence node together with that obstruction as a Belnap B -state rather than resolving it prematurely.

8.1 The problem space and the unconditional skeleton

Write `zauner_d = 2048` and `zauner_n = 11`, with the fiducial $B^{\oplus 11}$ the all- B Belnap word of length 11. The structural skeleton is inherited from `sic_povm_belnap_unconditional` of §4, instantiated at $n = 11$ and proved with zero `sorry`s:

$$|O_{B^{\oplus 11}}| = 4^{11} = 4,194,304,$$

$$\text{frobInner}(B^{\oplus 11}, g \cdot B^{\oplus 11}) = 22 \quad \text{for every } g \in \text{WHIdx}(11).$$

The orbit has the SIC cardinality d^2 , and the join-equiangularity constant $2n = 22$ is uniform across the orbit. No arithmetic hypothesis is used for either statement.

8.2 The transport map and its reduction

The forward transport assigns a complex amplitude to each of the 2048 coordinates from the Hamming weight of its bit vector. For $k \in \text{Fin } 2048$ with bit vector $v \in \{0, 1\}^{11}$ of weight $|v|$,

$$\varphi(B^{\oplus 11})(k) = \frac{\omega_d^{|v|}}{\sqrt{d}}, \quad \omega_d = e^{2\pi i/d}.$$

Theorem 8.1 (`transport_preserves_norm_sq`). $\varphi(B^{\oplus 11})$ is a unit vector: $\text{wh_normSq } 2048 \varphi(B^{\oplus 11}) = 1$.

Proof. Each amplitude has modulus $1/\sqrt{d}$, because ω_d lies on the unit circle: its exponent $2\pi i/d$ is purely imaginary, so $\|\omega_d\| = 1$ and $\|\omega_d^{|v|}\| = 1$. Hence every coordinate contributes $|\cdot|^2 = 1/d$, and the $d = 2048$ coordinates sum to $d \cdot (1/d) = 1$. Proved in `LEAN 4`. $\square$

The reduction $\psi: \mathbb{C}^{2048} \rightarrow B^{\oplus 11}$ reads a Belnap marginal at each register i by probing two representative coordinates: index 0 (bit i clear) and index 2^i (bit i set). Both carrying amplitude reads B ; only the clear probe reads T ; only the set probe reads F ; neither reads N . This is the structural content of the reverse morphism `AREV`, the read-back half of the split/fuse pair of §4.

Theorem 8.2 (`reduction_recovers_fiducial`). $\psi \circ \varphi = \text{id}$ on the fiducial: $\psi(\varphi(B^{\oplus 11})) = B^{\oplus 11}$.

Proof. Every transport amplitude is nonzero (a unit phase times $1/\sqrt{d} \neq 0$), so at every register both probes fire and the marginal reads B . The result is the all- B word $B^{\oplus 11}$. Proved in LEAN 4. $\square$

Theorems 8.1 and 8.2 exhibit φ and ψ as a transport/reduction pair realizing the Frobenius adjoint half $\mu \circ \delta = \text{id}$ concretely in $\mathbb{C}^{2048}$: the skeleton's structural content survives the round trip into the Hilbert space and back.

8.3 The existence ring and the conditional reduction

The arithmetic geometry mirrors $d = 12$. The SIC discriminant is $m_{2048} = (d-3)(d+1) = 2045 \cdot 2049 = 4,190,205 = 3 \cdot 5 \cdot 409 \cdot 683$, squarefree, so $F_{2048} = \mathbb{Q}(\sqrt{4190205})$ is real quadratic and the non-square hypothesis holds by `native_decide`. Over it sit the ray class field K_{2048} , the Stark unit ε_{2048} , and the 2048 complex embeddings, all instantiated from §3 at $d = 2048$.

Theorem 8.3 (`zauner_transport_theorem`). *If the mixed-signature Stark conjecture holds for K_{2048} , then `SICPOVM_Exists 2048`.*

Under the same hypothesis the structural join-equiangularity constant 22 lifts to the metric equiangularity $(d+1)|\langle \psi, D_{a,b} \psi \rangle|^2 = 1$, i.e. $|\langle \psi, D_{a,b} \psi \rangle|^2 = 1/2049$, reconciling the discrete skeleton with the $\mathbb{C}^{2048}$ frame (the fuse morphism `FFUSE` of the same pair).

8.4 The character obstruction

The reduction to Stark is the *true* branch. The *false* branch is a genuine obstruction to staying real. We prove it exactly, and state its scope precisely.

Theorem 8.4 (`evalf_character_obstruction`). *For $d = 2^n$ with $n > 1$, the Weyl–Heisenberg phase character $\omega_d = e^{2\pi i/d}$ is not real: there is no $r \in \mathbb{R}$ with $\omega_d = r$.*

Proof. $\omega_d = \exp((2\pi/d)i)$ has imaginary part $\text{Im } \omega_d = \sin(2\pi/d)$. For $d = 2^n > 2$ we have $0 < 2\pi/d < \pi$, so $\sin(2\pi/d) > 0$ and $\omega_d \notin \mathbb{R}$. Proved in LEAN 4. $\square$

The content is that the displacement phase itself is irreducibly complex: a real-valued character of the acting group cannot carry it, so the four-valued false branch (a purely real representation of the WH cocycle) is obstructed, and the projective representation requires genuinely complex cohomology. We are deliberately precise about what this is and is not. It is the exact non-reality of the phase, which is what blocks the real branch. It is *not* the stronger claim that no real fiducial can generate a WH SIC in these dimensions; that statement is adjacent to the real-equiangular-lines bound and we do not assert it. Stating the true obstruction and declining the unproven one is the honest form of the false branch.

8.5 The dialectic B -state

The two branches do not cancel. The true branch delivers existence under Stark; the false branch delivers the character obstruction; and both are theorems.

Theorem 8.5 (dialetheic_fiducial_paradox). *Assume the mixed-signature Stark conjecture for K_{2048} . Then*

$$\frac{\text{SICPOVM_Exists } 2048}{\text{true branch}} \quad \wedge \quad \frac{\neg \exists r \in \mathbb{R} : \omega_{2048} = r.}{\text{false branch}}$$

Recall the Belnap fixed point $\neg B = B$ of §4: the value B absorbs negation and holds true and false at once. Theorem 8.5 places the $d = 2048$ fiducial exactly there. The two dimensions the paper closes therefore sit at opposite Belnap values. At $d = 12$ the existence node is discharged and the fiducial reads T . At $d = 2048$ the existence node is open while its obstruction is proved, and the fiducial reads B : not a gap to be apologized for, but the correct four-valued type of an open problem whose skeleton and whose obstruction are both already theorems.

8.6 Proved unconditionally (0 sorries)

- **Existence of a SIC-POVM in dimension 12** (`crystal_forces_d12_sic`): exact fiducial constructed in the 2048-dimensional ring R , all 143 overlap identities and the norm machine-checked, transferred to $\mathbb{C}^{12}$ along an explicit homomorphism. Depends on no axiom of this development (§7, audit in §7.6).
- The Belnap four-valued lattice: $\neg B = B$, no explosion, Frobenius closure $\mu \circ \delta = \text{id}$ on $\mathbf{B}_4$.
- Product-lattice orbit cardinality: $|O_{B^{\otimes n}}| = 2^n$ (`whOrbit_card_eq_pow2`).
- The orbit gap: $2^n < 4^n$ for $n \geq 1$ (`product_lattice_orbit_is_insufficient`).
- The group bifurcation: $|\text{WHIdX}(n)| = 4^n$ (`group_bifurcation_lemma`).
- The main equivalence: `hilbert_embedding_equiv_zauber` by `rfl`.
- The Frobenius closure, join-equiangularity, and Born rule in the Belnap metric (`sic_povm_belnap_unconditional`, 22 theorems).
- The conditional existence theorem (`sic_povm_exists_via_stark`): $\text{Stark} \Rightarrow \text{SIC}$, from the axioms.
- **The $d = 2048$ transport** (`ZauberTransportMap`, 0 sorries): the unit-norm forward transport (`transport_preserves_norm_sq`), the reduction with $\psi \circ \varphi = \text{id}$ (`reduction_recovers_fiducial`), the conditional existence (`zauber_transport_theorem`), and the character obstruction that ω_{2048} is non-real (`evalf_character_obstruction`), with the dialetheic B -state conjoining the true and false branches (`dialetheic_fiducial_paradox`), Theorems 8.1–8.5.

8.7 Axiomatized with known mathematical content

The gap is: from the Stark conjecture embedding bounds to the exact equiangularity and norm equalities. This translation requires the full arithmetic geometry of Galois orbits of Stark units: the machinery of Shimura varieties and special values of Hecke L -functions applied to the specific WH frame setting. It is not a gap in the formalization strategy; it is a gap in mathematics.

None of these axioms enters the $d = 12$ existence theorem. The audit of §7.6 shows that `crystal_forces_d12_sic` depends only on LEAN 4’s standard foundations and the compiler trust of `native_decide`. The table above is the honest ledger of the *general- d* chain, not of the $d = 12$ result.

Axiom	Mathematical content
zauner_correspondence	Galois–Zauner: WH orbit overlaps controlled by the Zauner automorphism acting on the Stark unit (Appleby et al.)
equiangular_from_stark_axiom	Stark bounds imply equiangularity: the full arithmetic geometry of Stark units acting on WH frames
norm_of_normalized_axiom	Stark bounds imply the normalized fiducial has the correct norm; same arithmetic geometry gap
stark_equivalence	SICPOVM_Exists $\leftrightarrow$ Stark unit existence (Appleby–Flammia–McConnell–Yard structural link; placeholder in formalization)

Table 1: Honest gap markers: axioms carrying the open mathematical content.

8.8 Open

The mixed-signature Stark conjecture for K_d (Axiom 3.3). The Zauner conjecture for $d = 2^n$, $n \geq 2$ (ZaunerConjectureAtPow2). These are the same problem, identified by Theorem 5.2.

For $d = 12$ this subsection is empty. Every claim this paper makes about dimension twelve is a machine-checked theorem. For $d = 2048$ exactly one item is open: the unconditional existence `SICPOVM_Exists 2048`. The transport, the reduction, the conditional reduction to Stark, and the character obstruction of §8 are all theorems; what is not proved is the single node they surround.

9 Discussion

The `rf1` proof of Theorem 5.2 is the main structural result. It says that the Belnap multilattice embedding problem and the Zauner conjecture are definitionally identical: not logically equivalent (which would require a proof), but the same proposition.

This is possible because both definitions reduce to `SICPOVM_Exists (2n)`, the existence of a $WH(2^n)$ -covariant fiducial in $\mathbb{C}^{2^n}$. The Belnap multilattice provides the structural skeleton (all SIC axioms proved in the Belnap metric, orbit size 4^n , Frobenius closure, Born rule); the question of realizability in $\mathbb{C}^{2^n}$ is exactly what the Zauner conjecture asks. One is the algebraic statement, the other the geometric one; the formalization reveals they are syntactically the same.

The Stark link (§3) extends this to number theory. The discriminant formula $m_d = (d - 3)(d + 1)$ is not a coincidence; it is the field whose Galois theory organizes the WH orbit. The Galois–Zauner axiom 3.4 is the precise statement of the Appleby–Yadsan–Appleby–Zauner observation that the order-3 element of the Galois group (the “Zauner automorphism”) controls the SIC overlaps. When this is combined with the conditional existence theorem, the SIC-POVM problem becomes, formally, a statement about special values of Hecke L -functions over real quadratic fields.

The present formalization does not prove the Zauner conjecture in general or Hilbert’s

Twelfth Problem. It does prove one dimension of Zauner’s conjecture outright: $d = 12$, by exact construction, machine-checked from the ring presentation down to the kernel (§7). Exact $d = 12$ fiducials were known in the literature; what is new is the formal existence proof, to our knowledge the first for any dimension in any proof assistant, and with it the deletion of the existence axiom from the development. The method deserves a remark of its own: the entire analytic content of the proof was compressed into a finite set of decidable planks (ring identities under `native_decide`, rational sign changes for the root bracket, divided-difference positivity certificates) plus a single bridge lemma, so the boundary between computation and analysis is drawn exactly where the kernel can stand on the computational side. A numerical fiducial became an exact ring presentation, and the ring presentation became a theorem; at no point does the finished proof cite a floating-point quantity.

For the rest, the formalization locates the open content with precision, identifies the three levels as a single chain, and proves everything in the chain that is not open. The remaining axioms are not placeholders for missing proofs; they are the problem.

10 Artifact statement

All modules cited above live in the `p4ramill` library of the `p4raker` repository (<https://github.com/umpolungfish/p4raker>). Frozen review snapshot: branch `crystalline/manuscripts3-2026-07-07`, tag `crystalline-manuscripts3-v1` (commit `eea2c0c`); `main` continues independently. This manuscript and its companions `witness_vessel` and `chrysopoeia_2048` are frozen in `ig-docs` (<https://github.com/umpolungfish/ig-docs>) on the same branch name and tag.

11 Acknowledgements

The author would like to thank Harry T. Larson for imparting the importance of catching rising problems and never letting them go [5]. The present formalization is written in that spirit: the Stark–Zauner–Hilbert identification was not abandoned at the honest gap markers, and dimension $d = 12$ was carried from arithmetic prediction to a machine-checked existence theorem rather than left as a conjectural witness.

A The $d = 12$ fiducial primary coordinate

The machine-checked existence proof of §7 produces an explicit fiducial vector in $\mathbb{C}^{12}$. Its primary (zeroth) coordinate is a complex number whose real and imaginary parts are reproduced below to 1500 significant decimal digits, exactly as recovered by exact rational computation and transferred along the ring homomorphism of §7. This is the numerical witness against which the `native_decide` audit was run.

Real part

```
0.1062796056214234892401116553199052855065537600966583358463645722893226773031270658
298768906205486390530515471062546655918150810594007151620739921064457622344183907390
981527630502926525808781095296318291609375737019796692866006233984692833037108933368
```

557141825426546847294512590606360727608493535003406589845516102474605094572398156211
 791281798039542776561629744564194175203915687113788519889443755498620411265436001814
 914973032292646202106545011609706046505344547963396007168778128307754312885301096726
 162319934008918697128515617656809099978292761800100154277017013499197140932771280829
 474710901176233931186576815306983795289327943991170288570563450853952856813572224820
 323519766870258180444560783641518280663868403937579668795378352858216503948223011169
 697936687708120335054063539968448230213367085847359726613339750534761079437559558177
 170458770485092130024050377828296898248742359487812567890426747290383449758739584121
 891405313466683715232519962350717633557809679135595933926157129365473633858551371007
 959800932525205031235024239569645212695306698728965644437560979625978194688216215727
 988558883866587560774612330337201516160788444802262652227620504374101694536195129017
 480798390352751711845335405446315804414051104694360997639530534190402662133717527824
 453347700164869570420757011587658836186928728835106885477646747032721406405145380391
 458059143485376656502679909399982467453578747825974237188668321092959859046845380684
 589565644549976727349922297403872284018359670036895362355194841756494931895842349228
 42411

Imaginary part

0.0001630310010432694399587406315542496470085181880359789886468189132905265106516073
 242458412714148935119300482376377390198732003194349723579628890157953503274150184941
 922432970016102965159540029818814358845133768058607962230856300108004095482775591767
 607453478243239371216296563674622826993602940971440231503171851943962519739380612538
 628173386359275832075413408173617817250414210122534302281367413780218406525362813767
 377242250527245667716771638397748502815187010211458392191084019675813764757830592105
 763996740982160026003110762311735045878477445891538375462245767612445866581069506278
 477528999281835417095156724276857660199809330614562037038629851055255370186613991140
 255317589674458871903863720149657008412573166118787188346680142751797170677947903886
 52408286629791546693272534155441926719450493986023132122124022816540414743103338004
 194588521299511177701905727538444384985519664781080834434965132700067391423685777638
 814268440772310760076171105731976965738049432970937865573746731755532412624301987451
 469060456646790760478930383647270817656381589623592381926857668367406314975286153763
 506708672554330394678776563753044532036173045952393220064135915855846643399465208690
 514984645244764040372464298947848237029742729281882700217734069293190787967268119958
 187950201035331429126545813666316372019865413299603495502256362958288261801770979857
 300537231820251970793399646987127813352416035330544430067832933876151129206278709923
 794881789126474049505755088682279565026575170790581479272730707151904561896233556363
 8457817

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